



STRATEGY: PURCHASE

IMTRONICS TECHNOLOGY

Website: imtronics.in/pick-and-place-machine/
Address: G-312, Ground Floor, Sector-63, Noida - 201301
Email: sales@imtronics.in
Contact: 0120-4268884
Brand(s) Supplied: Imtronics

JB INSTRUMENTS PRIVATE LIMITED

Website: www.jbmicroscope.com
Address: J-9/36, UNIT NO:- 402 & 403, 2nd Floor, Rajouri Garden, New Delhi-110027
Email: info@jbstruments.com
Contact: 9873571533
Brand(s) Supplied: JB Microscope

KAMPFER AUTOMATION PRIVATE LIMITED

Website: www.kampferindia.com
Address: 119, E-20, BDA, Domlur 2nd Stage, Bengaluru, Karnataka - 560071
Email: sales@kampferindia.com
Phone: 080-25351440
Brand(s) Supplied: Tagarno

KEYENCE INDIA PVT LTD

Website: www.keyence.co.in
Address: Olympia Cyberspace, 10th Floor-Office No-1, No.21/22, Alandur Road, Guindy, Chennai - 600032, Tamil Nadu
Email: info@keyence.co.in
Contact: 1-800-103-0090
Brand(s) Supplied: Keyence



Zeiss Smartzoom-100
Supplied by: Carl Zeiss



KONTRELEK INSTRUMENTS

Website: dinolite-india.co.in/
Address: 44, Electronic Estate, Off Pune - Satara Road, Pune 411009
Email: dinolite@kontrelek.com
Contact: 9049997269
Brand(s) Supplied: Dino-lite

LAB INDIA INSTRUMENTS

Website: www.labindiastruments.com
Address: 201, Nand Chambers, L B S Marg, Near Vandana Cinema, Thane - 400602
Email: enquiry@labindia.com
Contact: 022 69086000
Brand(s) Supplied: Nikon

LABOMED INDIA PVT

Website: www.labomedworld.com
Address: 903, 9th Floor, Time Tower, M.G. Road, Gurugram India
Email: sales@labomedworld.com
Contact: 8882792899
Brand(s) Supplied: Labomed

LEICA MICROSYSTEMS

Website: www.leica-microsystems.com
Address: 3rd Floor B Wing Art Guild House Phoenix Market City LBS Road, Kurla West, Mumbai, 400070 India

WHEN TO USE WHAT

- Routine Production/Rework → Stereo or Digital Microscope
- Defect Localisation/Fine Features → Compound or Metallurgical Microscope
- Stress/Crystalline Material Studies → Polarising Microscope
- Mounted/Large Samples → Inverted Microscope
- Advanced Failure Analysis/Nano-scale Inspection → SEM

Features To Look While Selection

FEATURE	WHY IT IS IMPORTANT	WHAT TO LOOK FOR
Magnification Power	The core of PCB inspection lies in the ability to clearly visualise fine details like solder joints, micro-traces, and potential defects. Most standard inspections require magnification in the range of 10x to 100x, while high-density or multilayer PCBs may demand up to 500x-1000x for inspecting micro-components or BGA (Ball Grid Array) connections.	• Optical zoom vs digital zoom (optical preferred for clarity) • Zoom range adjustability (smooth, click-less control preferred) • Crisp edge-to-edge clarity without spherical or chromatic distortion
Working Distance	Working distance refers to the space between the microscope's objective lens and the PCB. A longer working distance allows ample room for soldering, rework, or probe testing without disturbing the focus.	• 70mm to 100mm is considered optimal for electronics inspection and hands-on work
Lighting	Clear visibility is essential to identify hairline cracks, poor solder joints, or oxidation. LED ring lights and oblique-angle lighting help reduce reflections from metallic surfaces and enhance contrast.	• Adjustable LED intensity (with dimmers) • Coaxial or side lighting for depth inspection • Polarised light filters to reduce glare • Choose microscopes with dual-lighting systems (top and bottom) for both opaque and translucent PCBs
Ergonomics	An ergonomically designed microscope reduces fatigue and eye strain, leading to faster and more accurate inspections.	• Eyepiece free design offering freedom of head movement and use of spectacles • Tilttable viewing heads or trinocular tubes • Adjustable height stands • Anti-fatigue posture design for prolonged sessions
Digital Connectivity	Modern digital microscopes streamline documentation, remote collaboration, and quality control. USB or HDMI output allows real-time display on monitors, and onboard cameras support high-resolution image and video capture.	• 1080p or 4K HDMI camera output • USB-C or USB 3.0 connectivity • Built-in software for image annotation and measurement • Wi-Fi/Cloud integration (for advanced collaborative workflows)